Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 03 · Part B: Mathematics of Signals and Systems Level 1

Track: Phasors, Linear Time-Invariant Systems, Convolution, Correlation, Deconvolution

Prerequisites: Problem 01 · **Estimated effort:** 10–14 hours

Convolution and LTI Systems: One Impulse Response Predicts Every Output

Almost every sensor, structure, amplifier and filter you will meet behaves, to good approximation, as a linear time-invariant system. Such a system is completely described by its impulse response, and its output to *any* input is a convolution. Build convolution from its definition, test systems for linearity and time invariance, identify a vibrating structure from a tap test, range a satellite with a spreading code by correlation, and learn why undoing a convolution is dangerous.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/03_convolution_and_lti_systems/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections,

and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

When an engineer taps a wing panel with a hammer and records the accelerometer, the recorded ringing is the panel's impulse response. Knowing it, the response to engine vibration, gusts or a shock load can be *computed* without any new experiment, because for a linear time-invariant (LTI) system the response to a sum of scaled and delayed impulses is the sum of scaled and delayed copies of the impulse response. That sum is the convolution. The same operation describes how a filter acts on data, how a telescope blurs a point source, and how a receiver correlates an incoming ranging code.

The mathematics of this problem (complex exponentials, convolution, correlation) is the foundation for everything that follows: the Fourier, Laplace and z transforms in problems 04–06 exist because convolution becomes multiplication in the transform domain. Do this problem slowly and verify each property numerically.

Theoretical Framework

1. Complex Exponentials and Phasors

Euler's formula $\mathrm{e}^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta$ lets a sinusoid be written $A\cos(\omega t + \phi) = \mathrm{Re}\{A\mathrm{e}^{\mathrm{j}\phi}\mathrm{e}^{\mathrm{j}\omega t}\}$. The complex number $A\mathrm{e}^{\mathrm{j}\phi}$ is the *phasor*. Sinusoids of the same frequency add by adding phasors:

$$\sum_k A_k \cos(\omega t + \phi_k) = \mathrm{Re}\left\{\left(\sum_k A_k \mathrm{e}^{\mathrm{j}\phi_k}\right)\mathrm{e}^{\mathrm{j}\omega t}\right\}. \tag{1}$$

Complex exponentials are the *eigenfunctions* of LTI systems: $\mathrm{e}^{\mathrm{j}\omega t} \to H(\mathrm{j}\omega)\,\mathrm{e}^{\mathrm{j}\omega t}$, which is why frequency response ($|H|$ and $\angle H$) fully characterises a sinusoidal steady state.

2. Linearity, Time Invariance, Causality, Stability

A system $\mathcal{T}\{\cdot\}$ is *linear* if $\mathcal{T}\{ax_1 + bx_2\} = a\mathcal{T}\{x_1\} + b\mathcal{T}\{x_2\}$, *time-invariant* if a delayed input gives an equally delayed output, *causal* if the output at n depends only on inputs at times $\le n$, and *BIBO stable* if every bounded input gives a bounded output. For an LTI system with impulse response

$h[n]$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = (x*h)[n], \qquad \text{BIBO stable} \iff \sum_{n} |h[n]| < \infty, \qquad \text{causal} \iff h[n] = 0\ (n<0). \tag{2}$$

Convolution is commutative, associative and distributive. For finite sequences of lengths N and M the result has length $N+M-1$ and costs $\mathcal{O}(NM)$ operations; it can be written as multiplication by a Toeplitz matrix built from h.

3. Second-Order Mechanical System

A structural mode with natural frequency ω_n and damping ratio $\zeta < 1$ has the impulse response

$$h(t) = \frac{1}{m\omega_d}\,\mathrm{e}^{-\zeta\omega_n t}\sin(\omega_d t), \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}. \tag{3}$$

Successive peaks of the ringing decay by the *logarithmic decrement* $\delta = \ln(x_k/x_{k+1}) = 2\pi\zeta/\sqrt{1-\zeta^2}$, so $\zeta = \delta/\sqrt{4\pi^2+\delta^2}$.

4. Correlation and Time-Delay Estimation

The cross-correlation $r_{xy}[m] = \sum_n x[n]\,y[n+m]$ peaks at the lag where y is a delayed copy of x. A maximal-length (m-)sequence $c[n] \in \{\pm 1\}$ of length $L = 2^m - 1$ has the periodic autocorrelation L at zero lag and -1 at every other lag, which makes it an ideal ranging code (the GPS C/A code is a length-1023 Gold code at 1.023 Mchips/s, one chip lasting about 0.98 µs, i.e. 293 m).

5. Deconvolution

If $y = h * x + n$ and $H(\omega)$ is small at some frequencies, the inverse filter $1/H$ amplifies the noise there without bound. A regularised inverse, $G(\omega) = H^*(\omega)/(|H(\omega)|^2 + \lambda)$, trades resolution for noise gain.

Data and Simulation Specification

Systems for Task 2	(a) $y[n] = x[n] + x[n-1]$; (b) $y[n] = x[n]^2$; (c) $y[n] = n\,x[n]$; (d) $y[n] = x[-n]$; (e) $y[n] = \lvert x[n]\rvert$; (f) $y[n] = \frac{1}{3}\sum_{k=0}^{2} x[n-k]$; (g) $y[n] = x[n] + 1$; (h) $y[n] = 1.1\,y[n-1] + x[n]$ (zero initial state).
Structure	Mode with $f_n = 12$ Hz, $\zeta = 0.03$, $m = 1$, sampled at 500 Hz, 4 s record; input: unit impulse, then random broadband force.
Ranging code	10-stage m-sequence ($L = 1023$) via `scipy.signal.max_len_seq(10)`; true delay $d = 337.4$ samples (fractional); chip-rate sampling then 4× oversampled; noise −10 to +10 dB per-chip SNR.
Deconvolution	Blur a pulse train with the structural response, add noise of 1% of the peak.
Real data	Tap a plastic ruler clamped over a table edge and record the smartphone accelerometer (Phyphox, highest rate) while it rings; use the recording as the measured impulse response in Task 3. Alternative: record a balloon pop with the microphone to obtain a room impulse response and convolve it with speech.

Your Task

1. **Convolution from scratch (Stage 2).** Implement the convolution sum with explicit loops, with a vectorised NumPy version and as a Toeplitz matrix product. Check against

`numpy.convolve` to machine precision, verify commutativity, associativity and distributivity numerically, and measure run time against N and M on log–log axes.

2. **Classify systems.** For each system (a)–(h) test linearity and time invariance numerically with random inputs and random shifts, decide causality and BIBO stability, and compute the impulse response for those that are LTI. Explain in one sentence why each failing system fails.
3. **Structural identification.** Simulate the structure with Eq. (3), extract ω_n and ζ from the ringing via the logarithmic decrement and via a nonlinear least-squares fit, and compare. Predict the response to a random force by convolution and check it against `scipy.signal.lsim`. Repeat the extraction on your real ruler recording.
4. **Phasors and eigenfunctions.** Combine three sinusoids of equal frequency into a single phasor and verify Eq. (1). Feed a sinusoid through a moving-average filter of length $M = 5$ and show that the steady-state output is the same sinusoid scaled by $|H| = |\sin(\omega M/2)/(M \sin(\omega/2))|$ and delayed by $(M-1)/2$ samples.
5. **Correlation ranging.** Verify the autocorrelation property of the m-sequence. Estimate the delay of the noisy received code by cross-correlation, refine it to a fraction of a sample by parabolic interpolation of the peak, and plot the RMS delay error versus SNR (200 trials per point). Compare with ranging using a single sinusoid.
6. **Deconvolution.** Recover the pulse train from the blurred noisy record with the inverse filter, then with the regularised inverse for several λ. Plot the RMS error against λ and explain the optimum.
7. **Stability.** Run system (h) with a bounded input and show the output diverges; compute $\sum |h[n]|$ for $h[n] = a^n u[n]$ as a crosses 1.

Self-check targets

Your convolution must agree with `numpy.convolve` to $< 10^{-12}$. Systems (b), (c), (d), (e), (g) are not both linear and time-invariant ((d) is linear but not time-invariant; (e), (b) are not linear; (g) is not linear because of the constant; (c) is linear but not time-invariant); (a), (f), (h) are LTI and only (h) is unstable. The periodic autocorrelation of the length-1023 m-sequence is exactly 1023 at lag 0 and -1 elsewhere. For $\zeta = 0.03$, $\delta \approx 0.189$.

Aerospace Application Link

A GPS receiver measures the code delay of each satellite by correlation. One chip is about 293 m of range, yet receivers reach metre-level code accuracy. Using your Task 5 results, plot the RMS ranging error (in metres, one chip = 293 m) against per-chip SNR for chip-rate sampling and for 4× oversampling with parabolic interpolation, and state the SNR needed for a 3 m error. Then estimate how the 1 ms code period limits the coherent integration gain.

Deliverables

D1 **Convolution notebook:** three implementations, property checks and timing plot.

D2 **System classification table:** linearity, time-invariance, causality and stability tests for (a)–(h), with the numerical evidence.

D3 **Structural identification:** impulse and random-force responses, parameter estimates by two methods, real ruler data.

D4 **Phasor and eigenfunction demonstration** for the moving-average filter.

D5 **Ranging and deconvolution:** delay-error curves with interpolation, and the regularised deconvolution error versus λ.

D6 **Memo (250 words):** "How accurate can code ranging be, and what limits it?" Cite at least two external sources.

Bonus (a) Implement fast convolution by overlap-add using the FFT and find the crossover length. (b) Show that a cascade of K identical moving averages approaches a Gaussian impulse response. (c) Estimate the room impulse response from a swept-sine recording and convolve it with speech.

Learning Resources

Signals and Systems

- Oppenheim, A.V., Willsky, A.S. *Signals and Systems*, Ch. 2 (LTI systems and convolution) and Ch. 3 (eigenfunctions).
- Lathi, B.P. *Linear Systems and Signals*, chapters on convolution and system response.
- Smith, S.W. *The Scientist and Engineer's Guide to Digital Signal Processing*, Ch. 6–7 (convolution, properties of convolution); `https://www.dspguide.com`.

Ranging and Correlation

- Kaplan, E.D., Hegarty, C.J. (eds.) *Understanding GPS/GNSS: Principles and Applications* (code correlation and the C/A code).
- Golomb, S.W. *Shift Register Sequences* (m-sequences and their correlation properties).

Tools

- SciPy `scipy.signal` (`lsim`, `max_len_seq`, `correlate`): `https://docs.scipy.org/doc/scipy/reference/signal.html`
- Phyphox: `https://phyphox.org`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.